

Energy storage integration with solar PV for increased electricity access: A case study of Burkina Faso

Hamza Abid ^a, Jagruti Thakur ^{a, *}, Dilip Khatiwada ^a, David Bauner ^{a, b}

^a KTH Royal Institute of Technology Stockholm, Sweden

^b Renetech AB, Sweden

ARTICLE INFO

Article history:

Received 10 June 2020

Received in revised form

6 April 2021

Accepted 13 April 2021

Available online 27 April 2021

Keywords:

Electricity access

Burkina Faso

Battery energy storage system

Smart energy systems

Grid extension

Photovoltaics

West Africa

ABSTRACT

Electricity access remains a challenge for the majority of the West African countries, wherein 5 out of 16 have an electrification rate of less than 25%, with Burkina Faso having only 9% of the rural population with electricity access in 2017. This study presents a techno-economic feasibility analysis of solar PV system integration with conceptualized Pumped Hydro Storage (PHS) and electric batteries for Burkina Faso. The study explores two cases (a) an off-grid PV with a storage system for rural areas and (b) a grid-connected PV system for an urban location. The least-cost configuration of PV with feasible storage is investigated using HOMER. The results show that Solar PV with PHS remains the optimal system configuration for both rural and urban cases even when the construction costs of lower and upper reservoirs are considered. Battery energy storage systems remain an economically expensive solution even when the added costs of pumped hydro storage are included, owing to the low lifetime and high capital costs of battery storage. The capital cost of PV remains to be the most dominating factor for both cases, signifying the importance of policy interventions for cost reduction of PV for increased green electrification in West African countries.

© 2021 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Burkina Faso is one of the least electrified countries in the world with overall electrification of 25% in 2017, wherein 65% of the urban and only 9% of the rural population is connected to the grid [1]. The objectives envisioned by the government are to reach an overall national electrification rate of 65% with an urban electrification rate of 95% and 50% for the rural case by 2030 [2].

The electricity generation capacity of Burkina Faso as of 2019 is around 300 MW and is dominated by fossil fuel-based (i.e. Diesel and Heavy Fuel Oil) generation (253 MW). Hydropower and solar generation capacity stood at 32 MW and 33 MW respectively [3] as shown in Fig. 1.

Despite the fact that Burkina Faso is located in one of the sunniest regions, the solar contribution to national electricity consumption in 2014 was only 0.8% [4], which rose to 5% with the addition of the 33 MW Zagtouli solar power plant to the grid in 2017 [5]. Burkina Faso depends heavily on electricity imports from its neighboring countries, hence the backbone of current national

policy is an extension of the 210 km long 225 kV interconnection [6], between Bolgatana (Ghana) and Ouagadougou (Burkina Faso), with a capital cost of 156 million USD. At present, only two 225 kV transmission lines exist in the national grid, which indicate a lack of grid infrastructure. As per 2017 JRC recommendations for Burkina Faso, the marginal cost of electrification could be reduced through the deployment of 374 MW of decentralized PV systems with an estimated cost of 1.7 billion euros to reach universal access to electricity by 2030 in Burkina Faso [4].

Similarly, a study by Bissiri et al. [7] considers the cost optimal electrification pathways in West Africa using a geospatial approach and highlights the needs of mini-grid PV systems for cost effectively serving the country's population.

Hence, along with the grid extension, there is a need to exploit the massive solar potential in the country. The country receives over 3000 h of direct sunshine per year [8]. In January 2018, the Ministry of Energy advertised plans to build eight solar parks with a capacity target of 100 MW [9]. Burkina Faso is one of the 15 member states of "The Economic Community of West African States" (ECOWAS), which is a regional organization of West African countries to promote economic integration among its members. ECOWAS in 2006 took the initiative to establish the West African

* Corresponding author.

E-mail addresses: jagrutithakur.ec@gmail.com, jrthakur@kth.se (J. Thakur).

Abbreviations

BGED	Breakeven Grid Extension Distance
COE	Cost of Energy
CRF	Capital Recovery Factor
ECOWAS	Economic Community of West African States
ESA	Energy System Analysis
FLES	Flat-Land Large Scale Electricity Storage
HFO	Heavy Fuel Oil
HOMER	Hybrid Optimization of Renewable Energy Resources
JRC	Joint Research Center
NPC	Net Present Cost
NREL	National Renewable Energy Laboratory
PHS	Pumped Hydro Storage
PV	Photovoltaics
WAPP	West African Power Pool

Power Pool (WAPP) to promote cooperation and integration of national power systems of fourteen inland countries in West Africa into a unified electricity market. ECOWAS and WAPP in 2012 approved a list of 59 priority projects for the region [6]. WAPP has also proposed several solar projects in Burkina Faso, one being the installation of three 50 MW solar PV plants in three different areas of the country by 2022 [10].

Energy storage is one of the key components of integrating intermittent renewables especially in the context of a weak grid infrastructure. The prices of Li-ion batteries fell around 18% from 2017 to 2018 [11]. The World Bank Group has committed one billion dollar to accelerate investments in the low-middle income countries under its program for “Accelerating Battery Storage for Development”. The main focus areas of the investment include hybrid solar PV with electric storage and mini-grids in low access areas [12].

Pumped hydro storage (PHS) is undoubtedly the most mature kind of storage technology, offering a lifetime of 25–100 years [12–15]. Overall, as per the IEA projections [14], the global PHS capacity is expected to increase by almost 20% (29 GW) in the next

five years whereas a total of 1.3 GW of PHS was added in 2016–2017 for the sub-Saharan African region [14].

According to “UNIDO World Small Hydropower Development Report 2016” [16], PHS represents around 10% of Burkina's electricity needs with two main hydroelectric power stations (Koumpenga, 14 MW and Bagre, 16 MW) totaling about 32 MW. In addition, also small-scale hydro plants Tourni (0.6 MW) and Niofila (1.68 MW) are in operation in Burkina Faso. The three major rivers in the country, Black Volta, Red Volta, and White Volta, offer an estimated potential hydro capacity of 138 MW. The longest one is the Black Volta (1352 km), located in the south-west. The two other principal rivers, the White Volta and Red Volta run from north to the south in the central plateau region. Planned future hydro projects are mostly located on the Black Volta with a capacity of around 80 MW [16].

The decreasing trends in the costs of electric storage, availability of water bodies, and vast solar potential in the country creates an opportunity for Burkina Faso to meet its electrification targets in a sustainable manner utilizing these resources.

This study aims to perform a techno-economic feasibility analysis of the integration of solar PV together with two storage options, viz. Li-ion batteries, and hypothetical PHS for electrification of Burkina Faso through different configurations. It aims to identify the optimal configuration of solar PV with storage for rural and urban cases based on the least cost. The Net Present Cost (NPC)¹ and Cost of Electricity (COE)² are analyzed for selecting the optimal system configuration. A cost comparison in terms of investments in grid extension (the network expansion from the national power transmission system to the new communities) vs standalone systems (self-sufficient decentralized solar PV with storage systems) is performed.

The scope of the study is limited to only two storage options: PHS and Li-ion electric batteries. The actual geographical details and site-specific parameters are not considered. The two storage options are considered for comparison for sustainable PV integration.

The paper is organized as follows. In section 2, a review of the relevant literature on the configuration of the optimal system for hybrid renewables integration is presented. Section 3 describes the methodology of the study and the data used for the analysis. Later the results from the analysis are compiled in section 4, along with the sensitivity analysis for the selected variables. This section also lists some aspects to consider for future outlook and policy implications, and finally a conclusion is presented in section 5.

2. Literature Review

Integration of fluctuating renewable energy resources like solar PV is inherent with its own challenges i.e. intermittency and grid stability. Several concepts have been introduced in the literature to devise systems capable of 100% cost-effective renewable energy solutions. One approach is the Smart Energy Systems approach as outlined in Ref. [17].

The Smart Energy Systems (SES) concept as opposed to the mere traditional deployment of ‘Smart Grid Information and Communications Technology’ (smart meters etc.) takes a systems approach to develop affordable 100% renewable energy systems as an alternative to fossil-based energy systems. This is characterized by

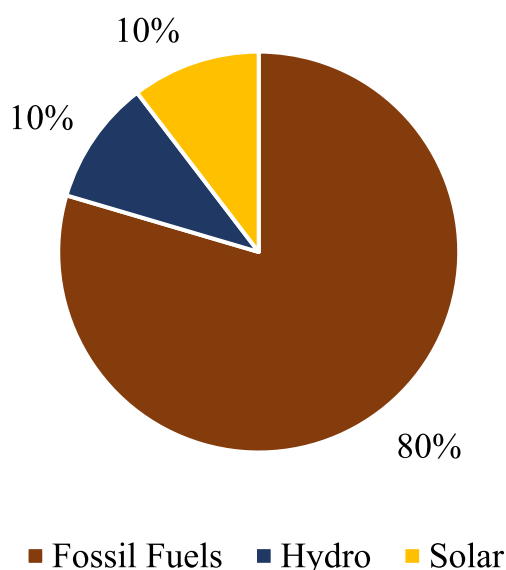


Fig. 1. Burkina Faso Electricity Generation Capacity [3].

¹ NPC is the present value of all the costs of installing and operating the components over the project lifetime, minus the present value of all the revenues that it earns over the project lifetime.

² COE is the average cost per kWh of useful electrical energy produced by the system.

synergies between different sectors of the energy system and utilization of storage technologies whilst limiting the use of bioenergy.

The storage technology depending on the application could be in the form of e-fuel production for transport or large-scale thermal heat storage for non-tropical climates. In smart energy systems, smart grid technologies are seen as part of the solution and not the whole solution [18] combined with different storage technologies. Pumped hydro storage is one of the cheapest and widely implemented forms of energy storage, making it a strong potential contender to pave way for future smart energy systems in tropical regions such as Burkina Faso. Similarly, owing to the continuous development in the battery storage technology, it is worthwhile to see how both of these storage technologies compare in their technical and economic potentials for the case of countries with vast solar potential and minimal electrification rates. Hence, this paper tries to perform a techno-economic analysis for the integration of both of these storage technologies for 100% solar PV renewable energy systems.

Several tools have been developed professionally for modeling these energy system elements. A review study performed by Conolly et al. [19] has looked into 37 such tools. The selection of a tool is highly dependent on several factors namely accessibility, the scope of analysis, availability of resource data, scenario timeframe, etc. One of the most widely used tools for simulation, optimization, and sensitivity analysis of the grid-connected and off-grid systems is HOMER, which has been ranked second in terms of the number of users by Ref. [19]. A systematic review study by Trotter et al. [20] on electricity planning and implementation in Sub-Saharan Africa lists HOMER as the most popular tool for sub-national level planning in Sub-Saharan Africa. The software has been extensively used in many studies for the cost comparison of investment decisions in Africa some of which are presented later in this section.

Similarly, a review of 19 software tools on hybrid renewable energy systems and sizing optimization [21] namely HOMER, Hybrid2, RETScreen, iHOGA, INSEL, TRNSYS, iGRHYSO, HYBRIDS, RAPSIM, SOMES, SOLSTOR, HySim, HybSim, IPSYS, HySys, Dymola/Modelica, ARES, SOLSIM, and HYBRID DESIGNE, lists HOMER as the widely used tool for optimization and sensitivity analysis with maximum combination of renewable energy systems among the 19 listed above.

HOMER reduces the complexity by providing access to a worldwide database for solar PV irradiation, the ability to design and choose industrially available components, and the option to model PHS as a storage type along with other technologies [22]. Hence, making the software a well-suited choice for this study as well. The literature on energy storage integration is dominated by optimal design studies with the purpose to identify the optimal size of storage with investment as a primary concern. Several studies have been performed to model energy storage systems for exploring optimal cost benefits. The following text lists some of these studies and describes how HOMER has been previously used for energy systems modeling in the literature.

In [23], G. Shrestha and L. Goel, have performed simulations for solar panel size and lead-acid battery size to optimize the operation of a stand-alone PV system, measured in terms of loss of load hours, energy loss, and total cost. Deb Chattopadhyay et al. [24] highlights the key issues to consider for deploying battery storage in developing countries with nascent electricity markets. It states that for battery storage to be price competitive in the developing world, “stacking of benefits and revenue streams” is needed. However, sizing is an important aspect when it comes to deploying these energy storage solutions. A study was performed by Nfah and Ngundam [25] on a hybrid configuration of pumped hydro, biogas, PV, and a battery system for a village in Cameroon. The optimal configurations were determined and a cost of energy of 0.352

\$/kWh was proposed using the HOMER optimization model. Similarly, another study for a hybrid system design for remote areas in Ghana was performed by Adaramola et al. [26] to determine the techno-economic feasibility of a wind-diesel generator and a solar hybrid system. A comparison of the grid-connected and islanded operation of solar PV micro-grid system for the case study of Iraq was done by Aziz et al. [27].

Mensour et al. [28] combined a case study analysis for a 10 MW grid-connected PV plant using HOMER with a GIS-based evaluation for solar farms locations in Morocco. Sigarchian et al. [29] used HOMER to model the domestic sector of Kenya using a generation mix of PV/wind and biomass generation by using electric batteries as storage. A techno-economic and performance analysis using HOMER is performed by Ref. [30] for the residential sector simulating different dispatch strategies. Ma et al. [31] have used PHS for a commercial load to find a feasible configuration using mathematical models. Daniel et al. [32] have performed an experimental and economical study of PV/diesel hybrid systems without storage for off-grid areas for Burkina Faso.

The literature review indicates that electric batteries have been considered extensively for standalone off-grid applications, but very little modeling has been done with PHS. There are some clear advantages of PHS such as lower life cycle cost as compared to batteries [31].

However, all existing schemes for PHS utilize the natural gravitational elevation and relief for water storage making it highly site-dependent. Studies such as those performed by Conolly et al. [33] have attempted to locate potential sites for pumped hydro storage. To reduce the geographical limitations of PHS and to make them more widely accepted, a relatively new method for the proven PHS technology is found in the literature. FLES (Flat- Land Large Scale Electricity Storage) is a variation of PHS that uses artificially created underground reservoirs for storage. This technique though does have added costs of excavation and construction of reservoirs but provides some benefits over the conventional natural PHS system, such as the reduction of geographical limitation for natural elevation and minimum visual impact. As outlined by a study performed for the Netherlands [34], the feasibility of a 1400 MW FLES in the Southern Netherlands proved technically and economically feasible. Such a system could prove to be beneficial when combined with intermittent renewables such as solar PV or wind.

To the best of the authors' knowledge, there is no scientific literature on the integrated assessment of PV systems with storage options (either PHS or electric battery storage) in the context of Burkina Faso. In this study, NPC and COE of different combinations of PV, electric batteries, and PHS are estimated using HOMER. This techno-economic feasibility for the grid and off-grid operations is important at this point as the region strives to achieve its electrification goals.

3. Materials and Methodology

This section describes the methodology for the comparison of PHS and electric batteries for the optimization of the solar PV plus storage systems. HOMER is used for the techno-economic optimization of the system. HOMER is designed by National Renewable Energy Laboratory (NREL) for studying renewable hybrid systems and to assist in the comparison of different generation sources in terms of cost and technical parameters in a poly-generation system.

Fig. 2 shows the inputs and flow of data for the analysis adopted in this study. Initially, the energy resources (solar PV, electric batteries, and PHS) are modeled based on their techno-economic parameters listed in Table 4.

The meteorological data such as the geographic solar irradiation is exported from the NREL database and a time series input of load

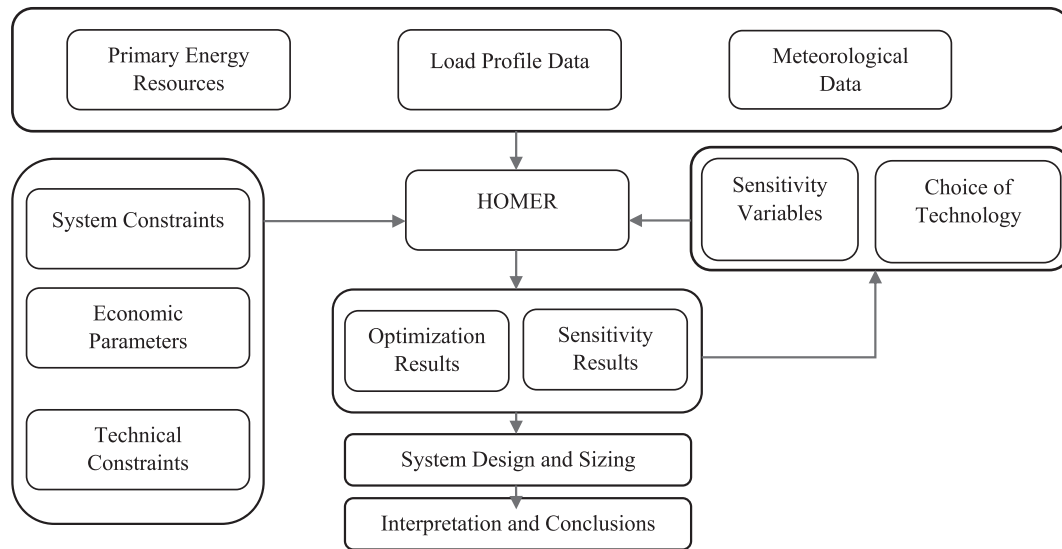


Fig. 2. Methodology flow of the study.

profile is fed into the system. The optimization of the system architecture in terms of the least cost and technical viability is performed. Sensitivity variables are identified based on the optimal configuration with the highest impact on the output. The system is then checked for robustness by varying these sensitivity parameters to assess the change in an optimal configuration. The off-grid PV system with storage mainly addresses the use of storage technologies with PV for rural applications such as remote villages. As the PV arrays are not grid-connected, a part of the study also deals with the economic comparison of grid extension break-even points and explores the following configurations:

- i) Solar PV + PHS
- ii) Solar PV + electric batteries

The grid-connected PV system with storage is studied for analyzing the feasibility of energy storage integration for the urban case.

Following configurations are modeled for grid-connected case:

- i) Grid-connected PV
- ii) Grid-connected PV + PHS
- iii) Grid-connected PV + electric batteries

The steps followed in the methodology (Fig. 2) are explained in detail in section 3.1 - section 3.4.

For performing the analysis, the load profile of the different consumers (rural and urban), as well as load profile for other energy usages, was determined.

3.1. Load Profile Assessment

The load profile analysis was performed for the following two cases:

- 1. Rural Population (Off-Grid)
- 2. Urban Population (Grid Connected)

The rural and urban population load profiles were further divided into residential and non-residential loads.

3.1.1. Urban and Rural Load Profiles

3.1.1.1. Rural Load Profile. To model the rural load profile in the country, a village (Sabou) located at the south-west of the country ($21^{\circ}12'06.01''$ N, $2^{\circ}22'26.1''$ W) in the Boulkiemde Province is considered. The major challenge in estimating the load profile in developing countries is the lack of quality data as extensively discussed by Girona et al. [5]. However, studies such as those done by Adeoye et al. [35] that provide modeling and forecasting guidelines for hourly residential demand profiles in West African countries provide close to realistic assumptions that can be used to model the residential urban and rural load demand profile. The relative error for the actual and forecasted electricity demand for 2013–2015 performed by Adeoye et al. [35] ranges from (- 7% to +2%). These models are typically based on behavioral, economic, social, and weather data to model the electricity demand for households. In this study, to model the hourly electricity demand, the results from the bottom-up analysis presented by Adeoye et al. [35] and the general parameters for social and economic factors of the respective rural and urban areas were mainly used.

• Residential Rural Load

The village has 45,000 inhabitants and 6429 houses [32] assuming seven inhabitants per household [5]. The per capita electricity consumption in the rural household was found to be 35 kWh per capita per year [35] as listed in Table 1. Fig. 3 shows the rural hourly residential electricity load profile for the village.

The peak hours are in the morning and in the evening, which can be due to a higher occupancy in the houses. This is simply the probability of having more occupants at the same time leading to the usage of lighting, electric fans, and water pumps [35]. The demand during the day remains relatively flat due to lower occupancy.

• Non-Residential Rural Load

Non-residential load consumption here is regarded as the load for societal use such as health centers, public lighting, etc. The parameters and assumptions considered for the social loads for a typical village in West Africa [5] are listed in Table 2.

Fig. 4 shows a typical daily non-residential load profile for the

Table 1
Data for residential rural load profile modelling

Parameters	Values
Population	45000 [32]
Number of inhabitants (per household)	7 [5]
Daily energy consumption pattern for households	1/3 energy consumed during daytime and 2/3 during evening and night [5]
Electricity consumption per capita	35 kWh per year [35]

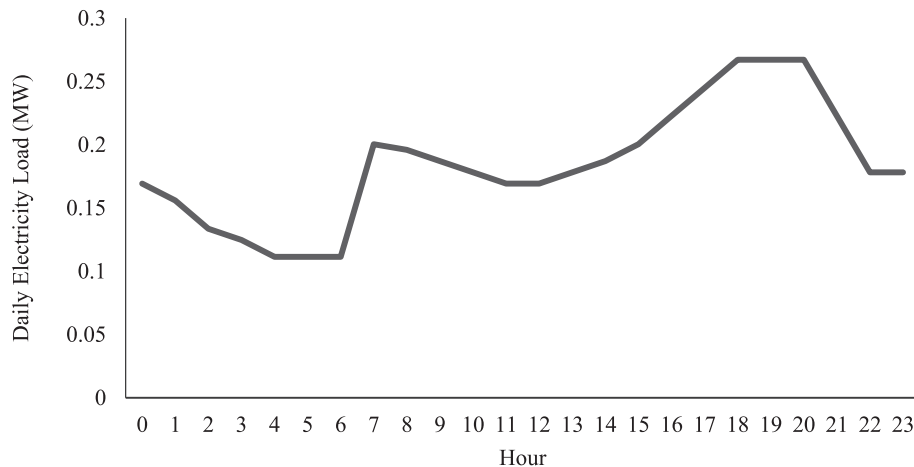


Fig. 3. Rural hourly residential electricity load profile [5,28, Authors' analysis].

village, the average consumption is 543 kWh/day and the peak load is 35.3 kW.

3.1.1.2. Urban Load Profile. The urban profile in this study was modeled for the city of Ouagadougou, the capital of Burkina Faso which has a population of around 2.2 million and is the country's largest city. The city receives an annual rainfall of about 800 mm which is spread over from May to October. The minimum average temperature in the cold season (December to January) is 16° Celsius. The maximum average temperature during the hot season (March to May) is about 43° Celsius [36].

• Residential Urban Load

The electricity consumption for residential consumers is mostly dominated by lighting (60%) followed by electric fans (20%) and air conditioner (10%), and the other miscellaneous consumptions are from washing and ironing in the year 2016 [35].

The load demand in different seasons (dry and wet) was considered constant due to minute changes in load demand between the two seasons. The seasonal variability analysis performed by Adeoye et al. [35] for 2016 suggests little difference in the demand profile of both seasons.

Hence, based on the general parameters mentioned in Table 3 and the limited variability of seasonal effects the hourly residential urban load profile was modeled as displayed in Fig. 5.

• Non-Residential Urban Load

The non-residential load in the urban profile consist of industrial, commercial and services demand sector as modeled by Ref. [35]. The overall non-residential load demand in the urban setting was synthesized using typical non-residential load profiles in the West African countries and the overall energy consumption [35].

The overall non-residential consumption in 2016 was found to be 720 GWh with a relative error of 1.1% [35]. The per capita non-residential demand was found to be 35 kWh per capita, and it is assumed that 2/3 of the energy is consumed during the daytime and 1/3 is consumed during evening and night.

The synthesized non-residential profile for Ouagadougou is as shown in Fig. 6, the average non-residential consumption is 201 MWh/day for a population of 2.2 million people with a peak demand of 13.5 MW.

3.2. System Configuration

The system consists of a residential load and a non-residential load connected to the PV array, storage devices (PHS or electric batteries), and a converter system. The individual components are explained in detail in 3.3. Fig. 7(a) shows the grid-connected urban system architecture where the DC components (electric batteries, PHS, and solar PV) are connected on the left whereas the AC grid and the AC residential and non-residential load are shown

Table 2
Data for non-residential rural load profile modelling [5].

Social Loads/Assumptions	Values
Social Center	2 kW
Health Center	5 kW
Hospital	15 kW
Electricity consumption pattern for social loads	2/3 energy consumed during daytime and 1/3 during evening and night

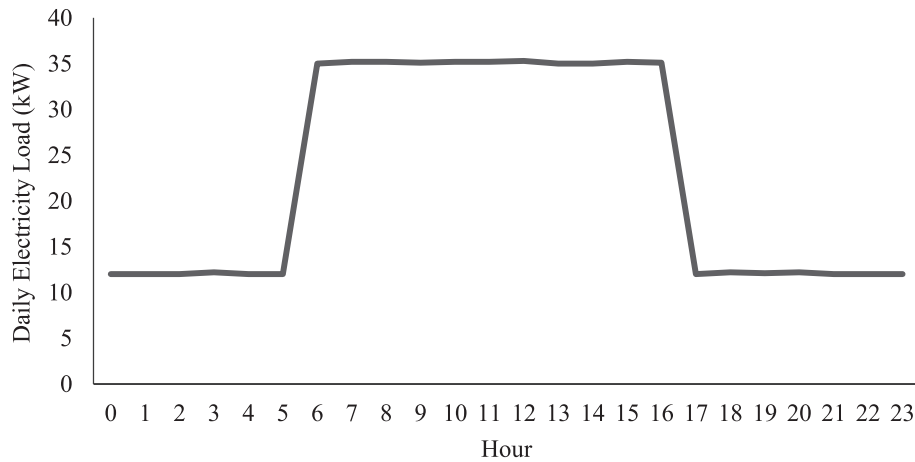


Fig. 4. Rural hourly non-residential electricity load profile [5,28, Author's analysis].

Table 3

Data for urban load profile modelling.

Parameters	Values
Population (Ouagadougou)	2.2 million [36]
Number of inhabitants (per household)	5 [5]
Daily energy consumption pattern for households	1/3 energy consumed during daytime and 2/3 during evening and night [5]
Electricity consumption per capita	45 kWh per year [35]

Table 4

Summary of input parameters.

Component	Capex	Operation and Maintenance	Lifetime	Replacement Cost	Efficiency	Source
PV System	850–2000 €/kW	10 €/kW/year	25 years	850–1500 €/kW	20.4%	[37,38]
AC-DC Converter	650 €/kW	0	20 years	600 €/kW	90%	[29,37]
Electric Batteries	200 €/kWh	5 €/kWh/year	10–12 years	200 €/kWh	90%	[39]
Pumped Hydro Storage	800 €/kW ^a	40 €/kW/year	25 years ^b	800 €/kW	81%	[13,40]

^a The capital cost here does not include excavation and construction costs for two new reservoirs which is discussed in detail in section 3.3.4.

^b The lifetime of the project is considered 25 years in this case, in reality it is 40–100 years for pumped hydro [13].

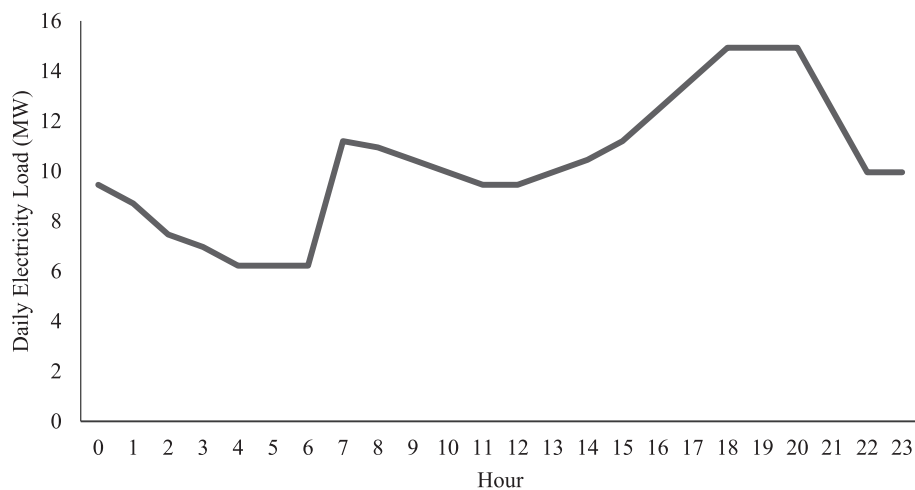


Fig. 5. Urban hourly residential electricity load profile [5,28, Author's analysis].

connected on the right side. Similarly, Fig. 7(b) shows the off-grid rural system architecture.

Based on the system architecture in Fig. 7, the system architecture for all the scenarios of urban and rural cases are designed in

HOMER. Fig. 8(a) shows the off-grid system architecture and Fig. 8(b) shows the grid-connected system architecture as modeled in the software. The specifications of the cost parameters are listed in Table 4 and values for load are considered based on the analysis

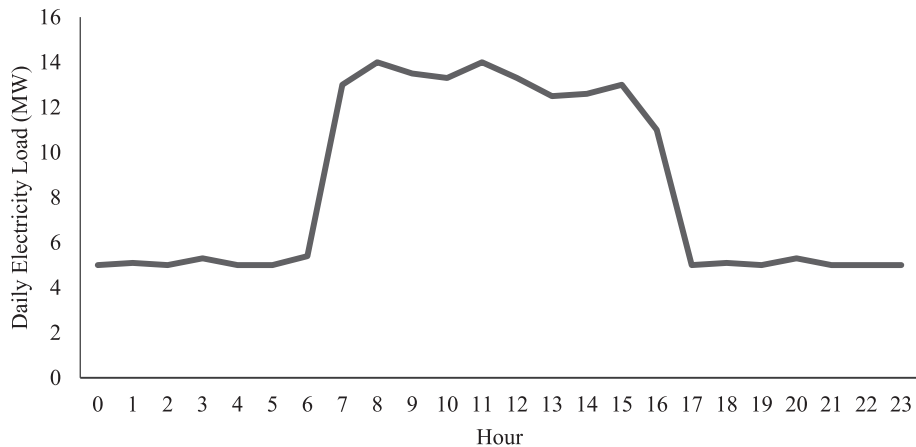


Fig. 6. Urban hourly non-residential electricity load profile [5,28, Authors' analysis].

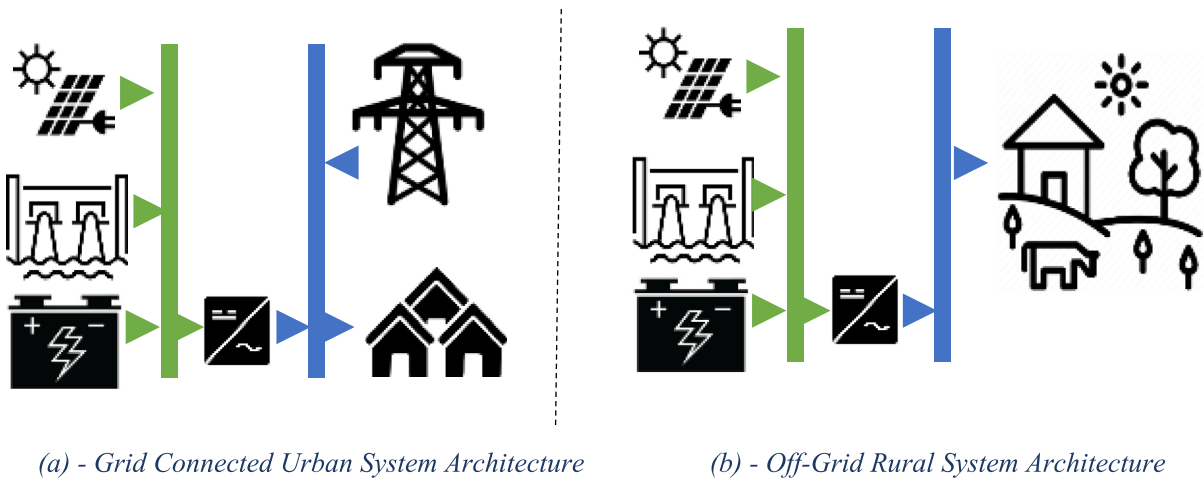


Fig. 7. (a) Grid connected urban system architecture, (b) Off-Grid rural system architecture.

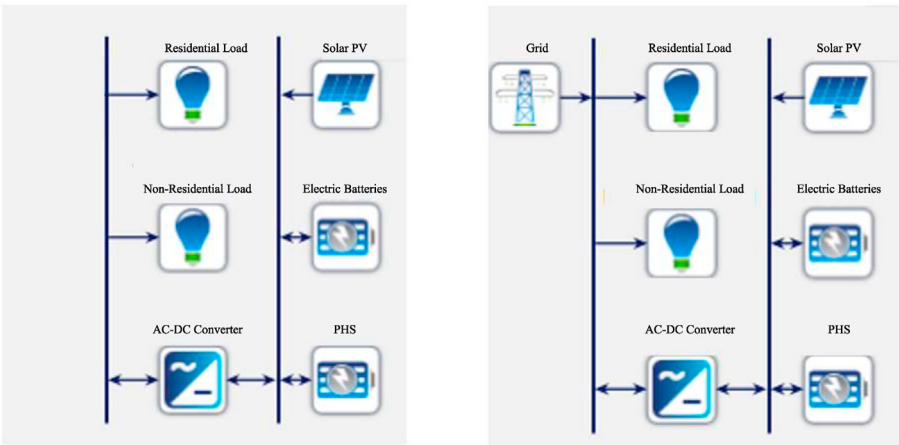


Fig. 8. (a) Off-grid system (HOMER), (b) Grid connected system (HOMER).

performed in 3.1.

3.3. Component Specifications

This section provides information about the renewable energy sources modeled in the system. The individual components are expanded in their respective subsections (Section 3.3.1 – Section 3.3.5), whereas Table 4 provides a summary of the parameters used as inputs.

3.3.1. PV System

The PV system considered for modeling was the SPR-E20-327-C-AC mono-crystalline module manufactured by Sun-power [22]. The cost range listed in Table 4, indicates the difference in capital costs of PV for community-scale and utility-scale PV. The lower value is for the utility-scale classified by 'Lazard's Levelized Cost of Energy' report 2018 [37] for the capacity of (>5 MW) which was used in the modeling for the urban case grid-connected analysis. The upper value is for the community scale system classified by Lazard's within the capacity of (1–5 MW) which was used for modeling the rural off-grid case in the analysis. Understanding that the price of PVs changes frequently, Lazard's report [37] was used as a reference and a sensitivity analysis was also performed to see the change in the optimal configuration for both urban and rural cases. To reduce uncertainty, a similar approach is used in the literature adopted by Sigarchian et al. [29]. There is a significant capital cost difference between the utility-scale grid-connected system and the off-grid system. The capital cost considered for the utility-scale PV is 850 €/kW whereas for a community-scale PV it is considered as 2000 €/kW as listed in Table 4. This difference is mainly due to the differences in the EPC (Engineering, Procurement and Construction) costs, developer overhead and net developer profit, all of which are termed as "soft costs" that are much lower for utility scale PV as compared to commercial scale PV as given by NREL cost benchmark Q1, 2018 report [41].

A derating factor of 0.95 was considered for the study and is used while evaluating operational efficiency of the panels in the analysis. The site-specific parameters such as the cost of land acquisition are not considered for this study.

3.3.2. Solar Energy Resource

The data values for the solar radiation were downloaded from NREL National Solar Radiation Database at the coordinates of 12.3714° N, 1.5197° W (Ouagadougou, Burkina Faso).

The annual solar radiation was found to be 5.64 kWh/m²/ day while the average clearness index was found to be 0.6, indicating a high potential of solar energy generation through PVs.

3.3.3. AC-DC Converter

To ensure continuity of flow of energy between the DC components (solar PV, electric batteries, PHS) and the AC components (grid, AC loads), a bi-directional AC-DC converter is used. The capital cost, replacement cost, and lifetime considered for a 1 kW system are 650 €/kW, 600 €/kW, and 20 years respectively [28,36].

3.3.4. Pumped Hydro Storage

PHS is used to store excess electricity from the PV arrays at times of peak generation and off-peak load demand. At times of a supply-demand imbalance, water is run through a turbine, producing electricity.

While PHS is a mature technology with a very long lifetime, low self-discharge, and a large storage capability [12–14], it does have its potential limitations. One of the major ones being the difficulty in identification of geographically suitable locations. Several

propositions have been proposed in the literature to tackle this: like the use of old coal mines as reservoirs or the use of FLES (Flat Land Large Scale Electricity Storage) [31,32]. FLES does not rely on natural relief and elevation to store gravitational energy but creates two artificial storage reservoirs. The proposition of PHS with no existing reservoirs is also proposed in the literature and cost estimates for excavation and civil works have been suggested. For example; a study by the "Oak Ridge National Laboratory, Sep 2016" [42], developed cost models for different cases such as the construction of new upper and lower reservoirs, refurbishment of upper and lower reservoirs, existing upper and lower reservoirs, etc.

According to Ref. [42], the costs associated with developing new capacity are 30 €/m³. This includes costs of leveling and grading, drainage, erosion control, soil excavation, rock excavation, and geomembrane. As mentioned in Table 4, the capital costs associated with PHS is 800 €/kW, if there exist two storage reservoirs with an elevation of 100 m and a storage capacity of 1000 cubic meters with a turbine efficiency of 81%. When additional costs of constructing new reservoirs (upper and lower) are included, the capital costs increase massively by more than 400% and the civil construction costs as a component of capital costs turn out to be almost 80%. The results are formulated considering that new reservoirs are needed to be constructed including the parameters listed in Table 4. The following parameters are not considered in this study that could potentially have an impact on the final costs.

1. Costs for construction of pipelines for the transportation of water from the available water bodies to the point of construction of PHS system in the case when two new reservoirs need to be constructed which could vary considerably based on the settlement considered for electrification.
2. Costs of land acquisition for the construction of PHS system.

3.3.5. Electric Battery Storage

Given the recent decline in battery prices, it is worthwhile to study the feasibility of comparison of batteries and PHS as storage. For this study, the economic parameters for 'In front of the Meter' battery applications for utility-scale were considered and are mainly based on the latest version of 'Lazard's Levelized cost of storage (LCOS)' report [39]. A single 100 kWh Li-ion battery unit was used in the model whose capacity scales up as per the demand input of the model. Based on the least cost principle in combination with solar PV, the number of batteries and the quantity of storage that satisfies the load demand without capacity shortages were determined. The capital cost, replacement cost, operation, and maintenance cost, lifetime, and efficiency are 200 €/kWh, 200 €/kWh, 5 €/kWh/year, 10 years, and 90% respectively [39].

3.4. Assessment Criteria

This section describes the main parameters on which different optimal configurations were compared for their feasibility. The main assessment criteria used in this study is the Net Present Cost (NPC) along with considerations for Cost of Energy (COE) and Breakeven Grid Extension Distance (BGED). Fig. 9 shows the flow chart of the simulation and optimization analysis for HOMER. After inputting the required data, different configurations with different component sizes are generated, the configurations that meet the required constraints are optimized based on minimum NPC. The desired optimal configurations are then sorted and compiled for further analysis.

3.4.1. Net Present Cost (NPC)

The objective function is the minimization of the total NPC. The

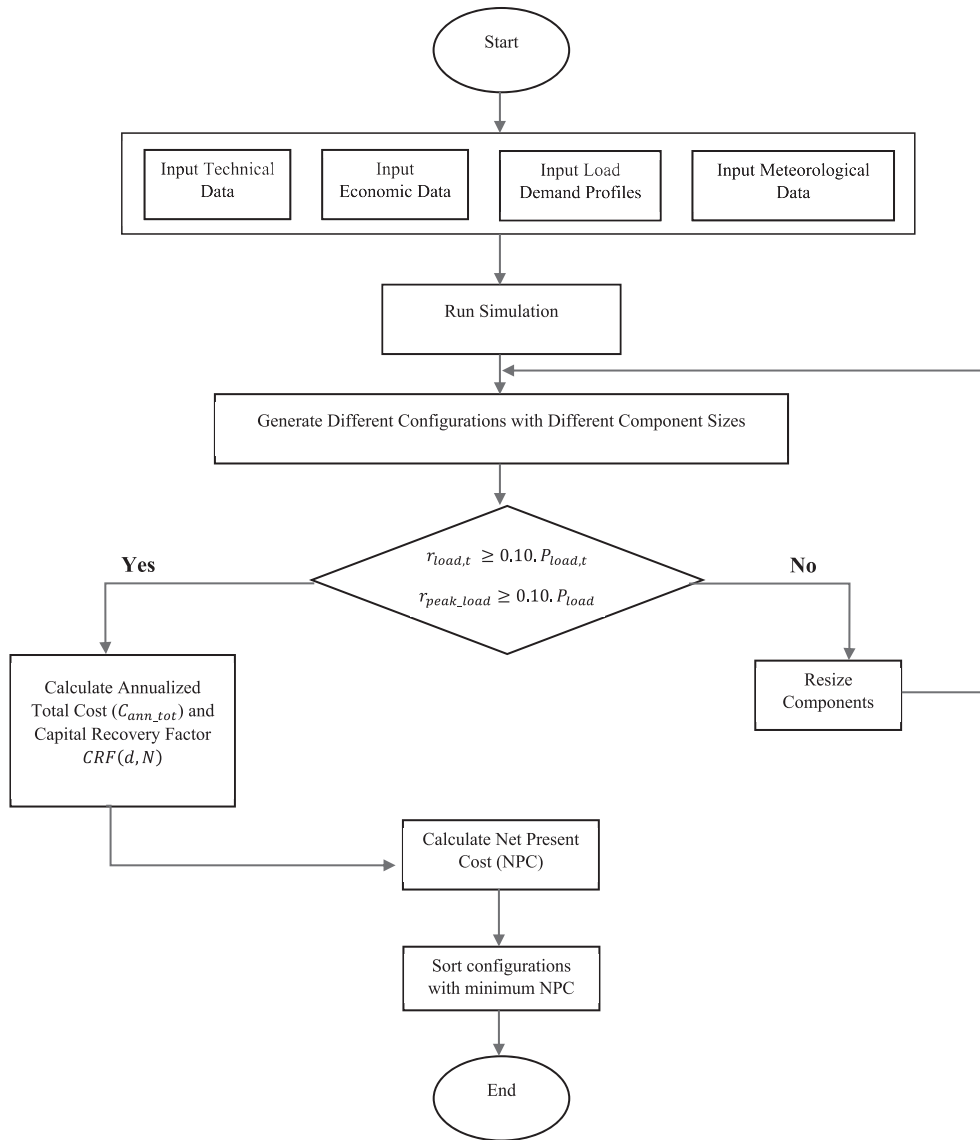


Fig. 9. Flow chart of Optimization Analysis using HOMER.

NPC is the present cost of all the system costs incurred over its lifetime minus the present value of revenues over its lifetime measured in euros. These costs include all capital costs, O&M costs, fuel costs, replacement costs, and costs associated with buying electricity from the grid. The revenues include the salvage value. The objective function in equation (1) is constrained by the inequalities (2)–(3). These constraints impose excess operating capacity to ensure the reliability of the system, expressed as a percentage of the variability of electric load in current time step (t) and annual peak load.

The model optimizes feasible system architectures for the given rural and urban cases. The feasible configurations are ranked in terms of NPC and the objective function is given as follows:

$$\min(C_{NPC}) = \sum_{components} \left[-R_0 + \sum_{t=0}^N \frac{R_t}{(1+d)^t} \right] \quad (1)$$

Subjected to the following constraints:

$$r_{load,t} \geq 0.10.P_{load,t} \quad (2)$$

$$r_{peak_load} \geq 0.10.P_{load} \quad (3)$$

Where.

R_0 is the initial investment [€]

N is the project lifetime

R_t is the net cash flow for each component (i.e., revenues minus costs incurred) [€]

d is the discount rate [%]

$r_{load,t}$ is the input operating reserve as a percentage of load in the time step t [%]

$P_{load,t}$ is the load in time step t [kWh]

r_{peak_load} is the input operating reserve as a percentage of the annual peak load [%]

P_{load} is the annual peak primary load

NPC is the present value of all the costs the system incurs over

its lifetime and is mathematically shown in equation (4) as follows:

$$NPC = \frac{C_{ann_tot}}{CRF(d, N)} \quad (4)$$

A Capital Recovery Factor (CRF) used in (4) converts a present value into a uniform annual cash flow series over the project lifetime (N) specified at a discount rate (d) as given by equation (5).

$$CRF = \frac{d(1+d)^N}{(1+d)^N - 1} \quad (5)$$

C_{ann_tot} is the total annualized cost of the system.

The annualized cost of a component is the cost that, if it were to occur equally in every year of the project lifetime, would give the same NPC as the actual cash flow sequence associated with that component. The total annualized cost of the system is as given in (6):

$$C_{ann_tot} = C_{ann_cap} + C_{ann_rep} + C_{ann_OandM} + C_{ann_fuel} - R_{ann_salv} \quad (6)$$

where C_{ann_cap} , C_{ann_rep} , and C_{ann_OandM} are the annualized capital, replacement costs, and operation and maintenance costs of all the components of the system, respectively. C_{ann_fuel} (since our system does not use any fuels, the cost of fuels for the generators is considered zero in our case). R_{ann_salv} represents the annualized total salvage value, that is the value remaining in a component of the power system at the end of the project lifetime [43]. The salvage value is calculated as follows:

$$R_{ann_salv} = C_{ann_rep} \cdot \left(\frac{N_{rem}}{N_{comp}} \right) \quad (7)$$

Where N_{rem} is the remaining life of the component in years at the end of the project which is given as:

$$N_{rem} = N_{comp} - (N - N_{rep}) \quad (8)$$

Where N is the project lifetime [years].

N_{comp} is the component lifetime [years].

N_{rep} which is the replacement cost duration in years is given as:

$$N_{rep} = N_{comp} \cdot INT \left(\frac{N}{N_{comp}} \right) \quad (9)$$

Where INT represents a function that returns the integer of a real number.

The discount rate in this project was considered as 5% and a lifetime of 25 years was considered for this analysis in accordance with the West African Power Pool recommendations [5].

3.4.2. Cost of Energy

Cost of Energy (COE) in €/kWh is defined as the average cost per kWh of useful electrical energy produced by the system [43]. The cost of energy is also an important assessment criterion for the study, given the very high cost of production of electricity in Burkina Faso. However, the objective function is based on the minimization of NPC and COE is one of the outputs of the model which can be used in combination with NPC to decide the most optimal configuration.

To calculate the COE, the annualized cost of producing electrical energy is divided by the useful electrical energy that the system produces per year.

$$COE = \frac{C_{ann_tot}}{E_{load}} \quad (10)$$

Where.

C_{ann_tot} is the total annualized cost of the system [€/year].

E_{load} is the total electric load served [kWh/year].

3.4.3. Grid Extension

Due to the very low electrification rate in the rural areas of Burkina Faso, grid extension break-even distance is an important assessment criterion for the feasibility of a standalone system for a rural population [4,44]. The break-even grid extension distance (BGED) is the distance from the system to the grid at which the NPC of extending a grid from the nearest available distribution line and the standalone operation of the PV plus storage is the same. BGED is calculated using equation (11).

$$D_{grid} = \frac{C_{NPC} \cdot CRF(d, N) - C_{power} \cdot E_{demand}}{C_{cap} \cdot CRF(d, N) + C_{OandM}} \quad (11)$$

where:

C_{NPC} is the total NPC of the stand-alone power system [€]

$CRF(d, N)$ is the capital recovery factor described in equation (5)

E_{demand} is total annual electrical demand [kWh/year]

C_{power} is the cost of power from the grid [€/kWh]

C_{cap} is the capital cost of grid extension [€/km]

C_{OandM} is the operation and maintenance cost of grid extension [€/year/km]

If the settlement is located beyond this distance from the grid, it would make economic sense to choose a standalone system for the electricity needs of that settlement. The required input parameters for calculating the break-even grid extension distance are the capital cost, O&M cost for the grid per km, and the grid electricity price. The existing distribution grid network of Burkina Faso has a length of 3550 km for 33 kV medium-voltage lines. There are plans to add another 6000 km of 33 kV lines to the existing network [5]. 'The Economics of Renewable energy Expansion in Rural Sub Saharan Africa' [45] gives an estimate of the capital cost for a 33 kV medium voltage line to be around 30,000 €/km or more. Also, since Burkina Faso faces constant capacity shortages, grid extension may trigger additional investments in the centralized system that would further increase the price of grid extension per km [4]. Considering this situation, the grid extension capital cost and O&M cost are considered as 40,000 €/km and 300 €/km respectively. The grid electricity price is provided by the 'Power Africa Burkina Faso Fact Sheet' [3] and is assumed as 0.1–0.3 €/kWh.

4. Results and Discussion

The results presented in this section are mainly divided into two parts, first, the grid-connected results for electrification are presented, including the optimization results, grid extension analysis, and sensitivity analysis. The optimization results showcase all the system configurations that are deemed feasible as per the constraints given in equations (2) and (3) and give the minimum NPC as stated in the optimization function (1). Later, the optimization results for the grid-connected urban load are discussed, followed by the sensitivity analysis for the same.

4.1. Off-grid

This section describes the results for the off-grid rural load with standalone PV and energy storage systems (PHS or electric batteries).

4.1.1. Optimization of System Architecture for Rural Consumers

After performing the simulation, a total of 1548 cases were simulated and 736 of them were found to be feasible, 812 were infeasible due to capacity shortage constraints and 360 were omitted for no sources of power generation. Ultimately, the two categories with the least NPC were considered for further analysis as listed in Table 5.

The two categories are (PV+ PHS) and (PV + Electric Batteries) listed as category 1 and category 2 respectively. The optimized PV capacity to meet an AC load of 1.81 GWh/year for category 1 is 1.6 MW with a PHS capacity of 16 MWh and an AC-DC converter capacity of 665 kW. The NPC breakdown by component for category 1 is shown in Fig. 10. The major portion of the cost is the capital cost of PHS since these calculations were obtained considering the civil construction costs of lower and upper reservoirs. The capital cost of PV also adds 4.2 million euros over a lifetime of 25 years which is around 42% of the total 10 million NPC of the system.

Fig. 11 shows the NPC breakdown of category 2. Here the major component is the capital cost of PV which is around 2.9 million euros. It is interesting to note that the COE in both cases is almost comparable.

An analysis was also performed to analyze the change in optimal configurations provided geographical availability of upper and lower reservoir without considering the cost of artificially developing it. The summary of the results is given in Table 6.

The NPC in the case of category 1 given the availability of reservoirs drops by 4 million euros as compared to the NPC when new reservoirs need to be constructed. Category 1 (PV+PHS) comes out to be the optimal configuration. The major contribution, in this case, comes from the capital cost of PV in both the categories as shown in Figs. 12 and 13.

4.1.2. Breakeven Grid Extension Distance (BGED)

As discussed earlier, BGED is an important criterion to assess the feasibility of a standalone system. For the standalone (PV+PHS) i.e. category 1 for a village population of 45,000, the distance is around 115 km as shown in Fig. 14.

The BGED for PV + Electric Batteries (category 2) is much higher i.e. 160 km as shown in Fig. 15.

These distances are calculated considering the costs of excavation of both upper and lower reservoirs. If these costs are neglected considering the geographic availability of upper and lower reservoirs the BGED is estimated to be 25 km as shown in Fig. 16.

It is observed from these results that it becomes increasingly more feasible to have a standalone system for remote areas that have geographical availability for a PHS system. In case both the reservoirs have to be constructed; the population needs to be a much greater distance (100 km–160 km) from the grid connection.

The sensitivity analysis in the next section analyses how this distance varies with the local population of the settlement which provides a general framework to assess the economic feasibility of standalone systems.

Table 5
Results of Off-grid scenario simulations.^a

Result Specification	Parameter	Unit	Category 1 (PV+PHS)	Category 2 (PV+ Electric Batteries)
System Architecture	PV	MW	1.6	2.2
	PHS	MW	1.4	0
	Electric Batteries	MWh	0	14.9
	AC-DC Converter	kW	665	906
Economic Specifications	NPC	€	10.1 Million	12.3 Million
	Capital Cost	€	9.53 Million	9.29 Million
	COE	€/kWh	0.4	0.4

^a The results are calculated considering the capital cost of construction for both reservoirs.

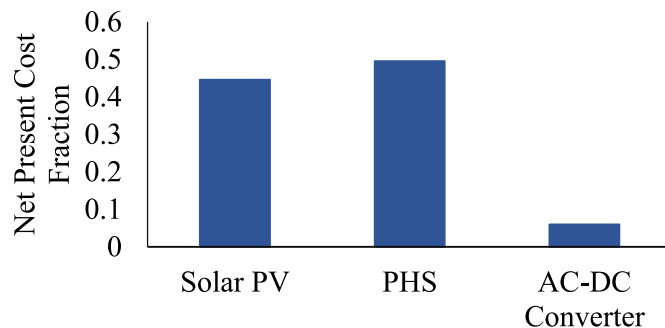


Fig. 10. NPC breakdown of category 1 (1.7 MW Solar PV and 1.4 MW PHS) by component with additional costs of excavation.

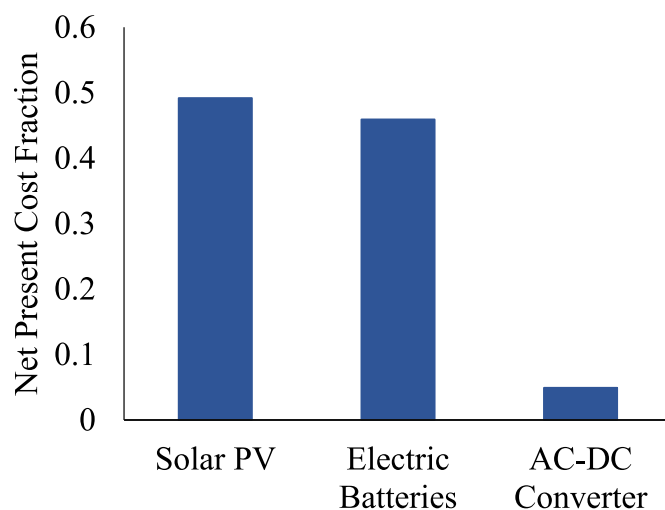


Fig. 11. NPC breakdown of Category 2 (2.2 MW Solar PV and 14.9 MWh Electric Batteries) by component with additional costs of excavation.

4.1.3. Sensitivity Results

Sensitivity analysis provides a good understanding of the robustness of a model. Given the continuous trend of decrease in PV prices and considering that the capital cost of PV is the major component of NPC in all the cases, a sensitivity analysis on the capital cost of PV prices was performed with cost multipliers of 0.25, 0.5, 0.75, and 1, and subsequently the effect was observed on the change in NPC and COE as shown in Fig. 17. The NPC of the project almost decreases by half with the decrease in PV Capex by 1/4th of the present price.

Fig. 18 shows how the NPC evolves with the change in demand. The NPC and PV capacity decrease exponentially with the decrease in demand.

A sensitivity analysis for the break-even grid extension distance

Table 6
Results of Off-grid scenario simulations.^a

Result Specification	Parameter	Unit	Category 1 (PV+PHS)	Category 2 (PV+ Electric Batteries)
System Architecture	PV	MW	1.6	2.2
	PHS	MW	1.4	0
	Electric Batteries	MWh	0	14.9
	AC-DC Converter	kW	523	906
Economic Specifications	NPC	€	6.15 Million	12.3 Million
	Capital Cost	€	5.54 Million	9.29 Million
	COE	€/kWh	0.25	0.4

^a The results are calculated considering geographical availability of both reservoirs without additional costs of excavation.

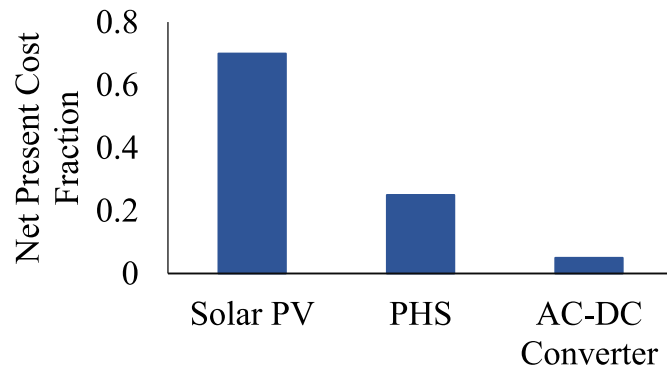


Fig. 12. NPC breakdown of Category 1 (1.6 MW Solar PV and 1.4 MW PHS) by component without additional costs of excavation.

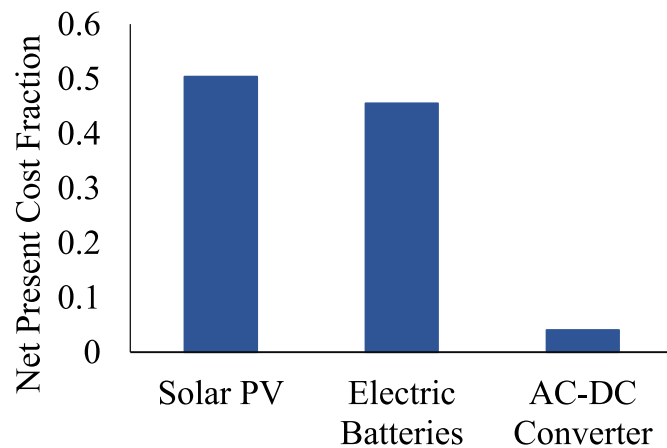


Fig. 13. NPC breakdown of Category 2 (2.2 MW solar PV and 14.9 MWh Electric Batteries) by component without additional costs of excavation.

was also performed with varying population sizes to analyze the feasibility of a standalone system in terms of cost for different sizes of rural settlements. This would give us insights into how BGED can be used to assess grid extension feasibility for varying rural populations. The optimal system with varying populations remains the same i.e. PV + PHS. Table 7 shows the residential load demand of a settlement with their corresponding population and BGED.

The results of the sensitivity analysis are displayed in Fig. 19. It is observed that it becomes more feasible to have a standalone system for a population of fewer than 6000 inhabitants that are at least 26 km away from the nearest grid when construction costs of PHS are not considered.

A similar approach could provide useful insights for other neighboring West African countries facing similar challenges of low electricity access.

4.2. Grid Connected

This section analyzes the results of the grid-connected utility-scale PV with storage options (PHS or electric batteries) for an urban load profile. First, the optimization results for the system architecture for the least NPC are presented followed by a sensitivity analysis of NPC and PV capacity size with changing PV capital cost.

4.2.1. Optimization of System Architecture for Urban Consumers

The grid-connected optimization results are compiled in Table 8. The results optimized are mainly divided into three categories based on least NPC.

Category 1 (PV+ PHS+ Grid) is the cheapest option of all. Category 3 (Grid Only) which has a high NPC because of the very high cost of grid electricity prices in Burkina Faso is also presented which justifies the need for PV production unit to drive down the cost of energy. It is interesting to note that NPC of the project decreases by a mere 8% from 352 million euros to 323 million euros by the addition of storage over a lifetime of 25 years.

Also, there is a higher renewable fraction³ of 88.3% as compared to 48.6% for category 2 since less energy has to be purchased from the grid to meet the load demand, and most of the energy generated on the grid in Burkina at present is through fossil fuels. Since for this model, the costs of land acquisition are not taken into account and there might be other costs associated with PHS that are site-specific, the high capital investment of 236 million euros for category 1 as compared to 80 million euros for category 2 might not be justified. However, on the other hand, given that the current COE in Burkina Faso is quite high, an addition of 71 MW of PV capacity (category 2) with an initial capital of 80 million euros, decreases the NPC by 25% as compared to category 3 (grid only).

Fig. 20 shows the NPC breakdown of category 1 system architecture which is the least NPC system. Like the off-grid situation, the main component in the costs is the capital cost of solar PV and PHS.

4.2.2. Sensitivity Results

A sensitivity analysis of the price of the PV was performed to see the effect on NPC and PV capacity. Fig. 21 shows the change in NPC and PV capacity demand with the change in capital price. It is seen that the least NPC system architecture remains the same i.e. category 1 (PV + PHS + Grid) and with the decrease in PV prices, more PV could be incorporated into the system that optimizes the cost and reduces the NPC of the system. This seems plausible as shown in Fig. 20 that the main component of NPC is the CAPEX price of the PV system.

The results presented above for both the rural and urban case depend considerably on the input parameters like population,

³ Renewable fraction is the fraction of the energy delivered to the load that originated from renewable power sources.

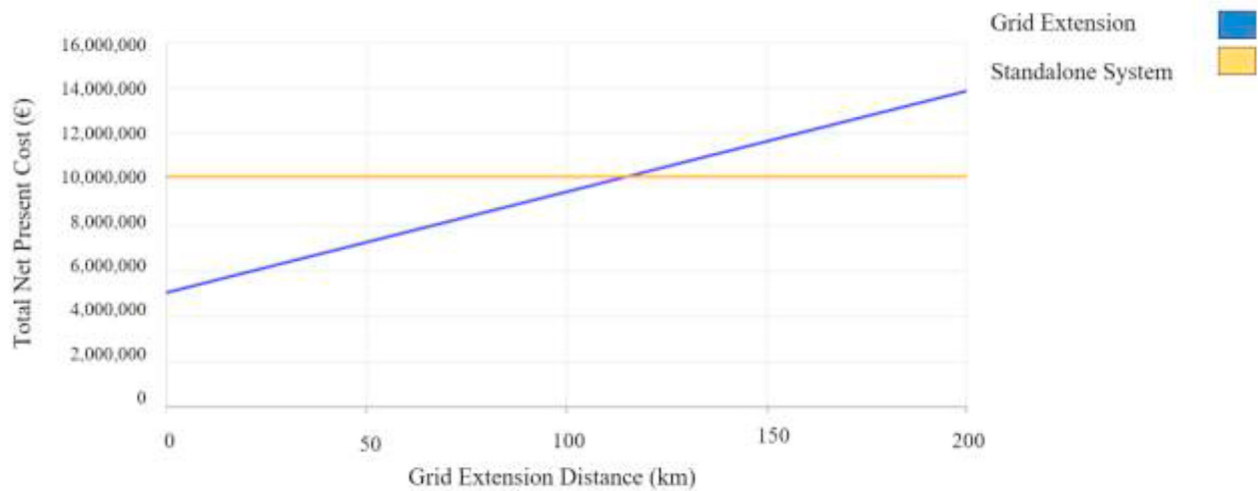


Fig. 14. BGED for Category 1 (1.7 MW solar and 1.4 MW PHS) with additional costs of excavation.

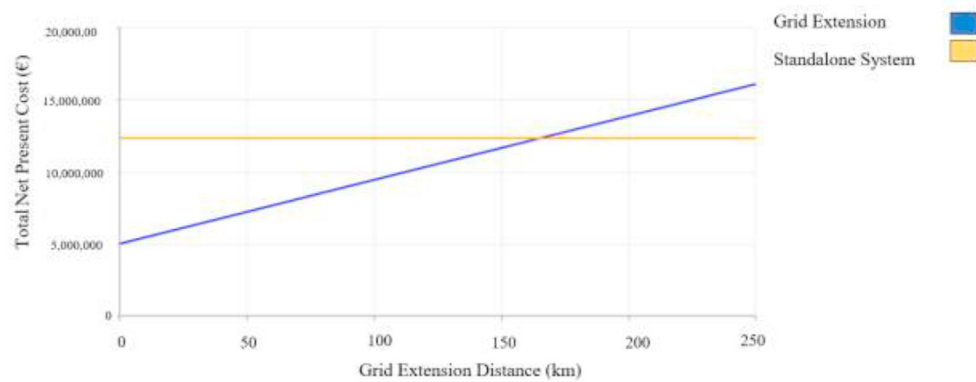


Fig. 15. BGED for Category 2 (2.2 MW Solar PV and 14.9 MWh Electric Batteries) with additional costs of excavation.

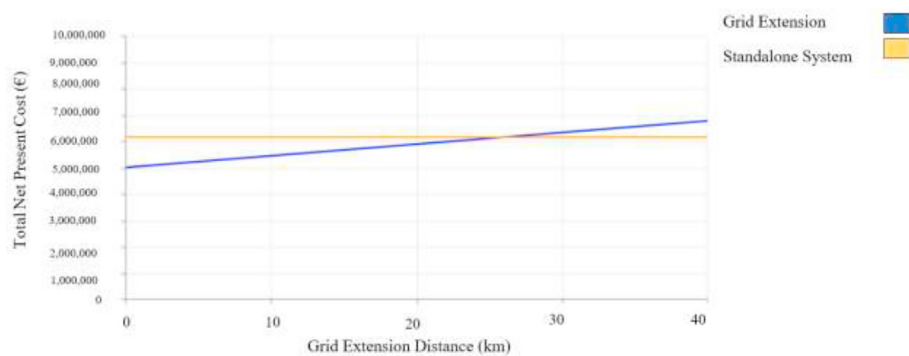


Fig. 16. BGED for Category 1 (1.6 MW Solar PV and 1.4 MW PHS) without additional excavation costs.

location, cost parameters, project lifetime, and system architecture.

This study is compared with other studies for observing the COE of different configurations of energy technologies performed for Burkina Faso and Sub-Saharan Africa. Table 9 shows that the range of COE is 0.25–0.69 €/kWh depending on the technologies considered for system configuration.

4.3. Future Outlook and Policy Implications

Additional studies need to be performed to address the limitations in this study. Future techno-economic analyses from studies such as the application of different techniques for locating prospective PHS sites such as GIS modeling [48] and computational

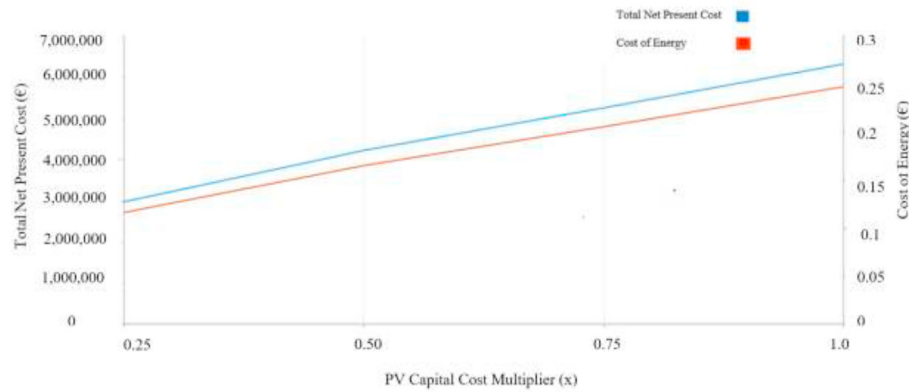


Fig. 17. Sensitivity of COE and NPC with respect to PV capital cost.

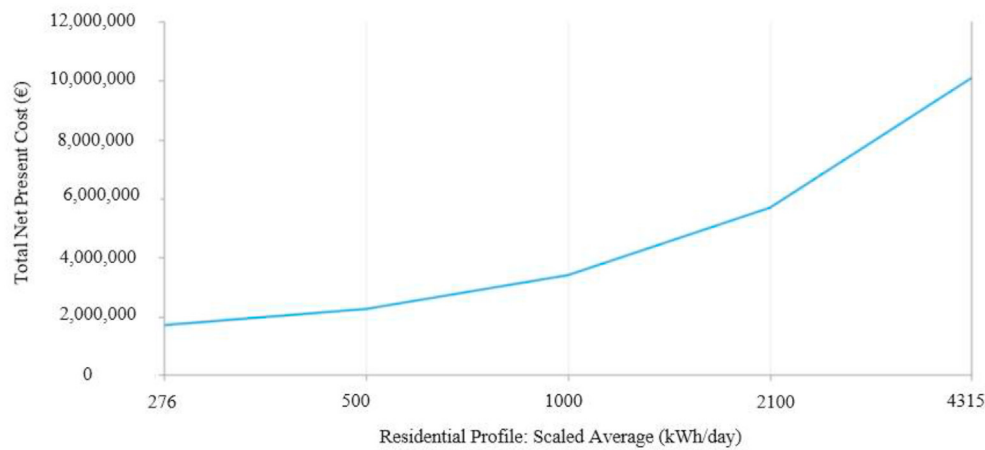


Fig. 18. Sensitivity of NPC with respect to residential profile.

Table 7

Average load demand with corresponding population and BGED.

Scaled Average Load Demand per settlement (kWh/day)	Population	BGED (km)
4315	45000	115.29
2100	22500	67
1000	11250	41
500	5625	26.63
276.375	2812.5	19.44

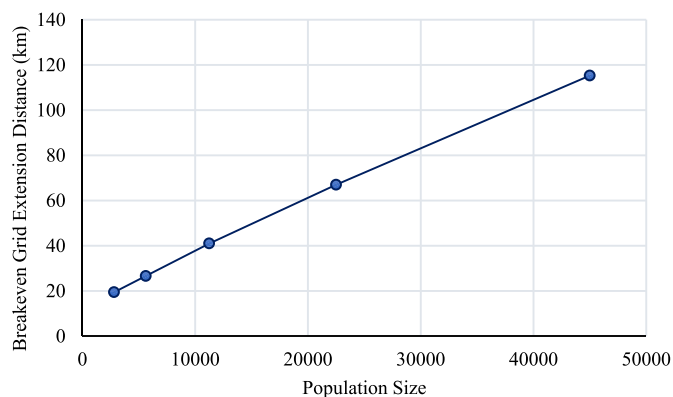


Fig. 19. Sensitivity of BGED by population size.

methods such as developed by Conolly et al. [33], when combined with the results of this study would provide useful insights into the role of PHS and FLES as a storage mechanism for off-grid systems for increased electricity access in Burkina Faso and energy planning in general. Further, this study uses COE and NPC as the main economic indicators for comparative feasibility, as these are traditionally used for assessing the different renewable energy systems. However while performing the comparison with the traditional fossil-based system, Energy System Analysis (ESA) cost method as outlined by ref. [49] would provide more useful insights as the ESA approach takes into account the whole energy system from value chain to distribution and internalizes external costs such as intermittency of renewables, environmental damages, cost of CO₂, etc. HOMER has been used in this study for sizing optimization of different storage elements for urban and rural cases. The deployment of different heuristic techniques like Particle Swarm Optimization (PSO) could provide better approximations for COE and

Table 8
Optimization results for grid connected simulations.

Result Specifications	Parameter	Unit	Category 1 (PV+PHS + Grid)	Category 2 (PV + Grid)	Category 3 (Grid Only)
System Architecture	PV	MW	136	71	0
	PHS	MW	26.9	0	0
	Batteries Li-Ion	MWh	0	0	0
	AC-DC Converter	MW	29	29.4	0
Economic Specifications	NPC	€	323 Million	352 Million	471 Million
	Capital Cost	€	236 Million	80 Million	0
	COE	€/kWh	0.129	0.1	0.2
	Renewable Fraction	%	88.3	48.6	0
	Grid Purchases	GWh	20	92	167

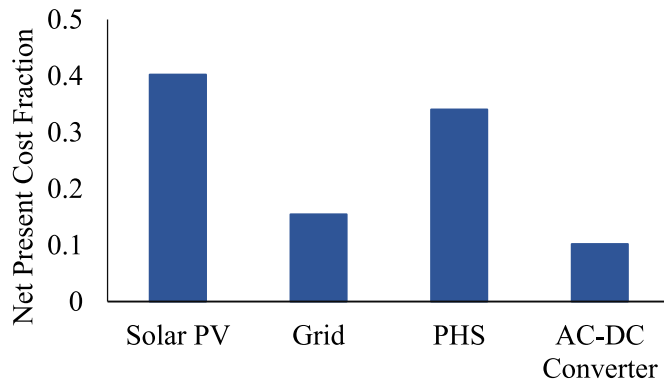


Fig. 20. NPC breakdown of Category 1 (136 MW Solar PV and 26.9 MW PHS) by component.

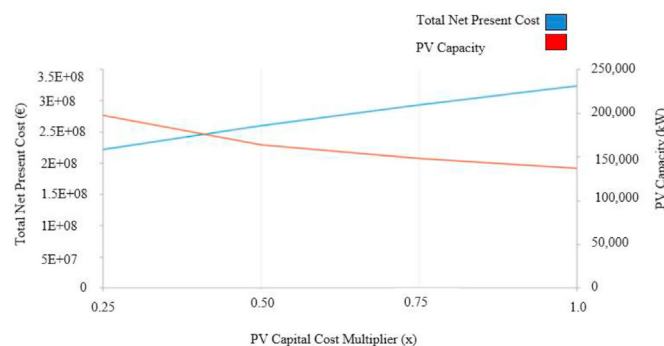


Fig. 21. Sensitivity analysis of PV capital cost on NPC and PV capacity.

NPC for both the systems as outlined by comparative analyses performed by Fodhil et al. [50] and Clarke et al. [51]. The results could also be refined further if better approximations of local load profiles and costs data are made available. The Breakeven Grid Extension Distance (BGED) results in this study provide useful insight for electrification of far off places in Burkina Faso and it would be interesting to see these results combined with the

aggregated information on scattered non-electrified settlements throughout the country and the costs comparisons between decentralized and centralized electrification on a national level.

Access to reliable electricity is a pre-requisite for economic growth and a lack or limited access to modern energy services could seriously hamper the development aspects of ECOWAS member states [52]. The present policy of Burkina Faso is mainly dominated by a potential increase in centralized generation capacity and grid extension [5]. This study suggests that along with strengthening the national grid, future policies should also focus more on decentralized standalone energy systems. In doing so, the rate of electrification could be increased at a lower investment cost. The potential barriers for its practical implementation are the high investment costs, as seen from the results both for the urban and rural case, the capital cost of PV is the most dominating factor. Driving down the capital costs of PV by shifting fossil fuel subsidies to solar PV and in-house, production could potentially lead to better outcomes. Burkina Faso is currently setting up a regulatory framework for the purchase of electricity from IPPs (Independent Power Producers) [53], rapid unbundling of the energy sector and effectively attracting private investments in storage along with subsidies favorable for financing PVs to reach the target of 65% electrification by 2030.

Energy system elements like storage especially pumped hydro storage should be a part of future policy design. The storage element not only reduces costs for electrification but also pave way for future synergies between different sectors such as transport, electricity, etc. in effect shifting the conventional fossil-based energy systems to a coherent smart energy system.

5. Conclusion

The study finds the need for extended solar penetration in Burkina Faso in response to the challenges of low electrification rates in the country. Electric batteries remain an expensive solution i.e. the NPC for PV with batteries is more than double the NPC of PV with PHS configuration, even when new reservoirs need to be constructed, the high capital cost of batteries and a low lifetime as opposed to a very long lifetime of PHS and FLES make batteries an expensive option to consider with the present costs. PV plus PHS option is deemed a much feasible option in terms of NPC especially

Table 9
COE comparison of different studies.

Study	System Architecture	COE (€/kWh)	Location
Our study	Off-grid Solar PV + Electric Batteries	0.4	Burkina Faso
Our study	Off-grid solar PV + PHS	0.25	Burkina Faso
Bachir I. Ouedraogo et al. 2014 [46]	Off-grid Solar PV + Diesel Generator	0.45	Burkina Faso
Bachir I. Ouedraogo et al. 2014 [46]	Standalone Diesel Generator	0.69	Burkina Faso
Sara Ghaem Sigarchian et al. 2015 [29]	Off-grid solar PV + Wind + Biogas Generator	0.25	Kenya
Nasser Yimen et al. 2018 [22]	Off-grid solar PV + bio-gas generator	0.256	Northern Cameroon

given the availability of reservoirs. It is observed that for the grid-connected systems, adding storage reduces the NPC of the system by 8% as compared to grid-connected PV. The costs of deploying storage could be further reduced if future policies allow mechanisms such as energy arbitrage that allow stacking of energy storage benefits and revenues. However, the study finds a definite need to add more solar energy to the present electricity mix of Burkina Faso. It is found that adding grid-connected solar PV in an urban setting could reduce the cost of energy by half. It would also increase the renewable fraction by more than 48%. The sensitivity analysis indicates that driving the capital costs of PV has the most positive effect on driving down the NPC of both urban and rural cases. A support system that complements the capital cost of PV would prove to have the most effect on lowering the costs of electrification in the region. It is understood that access to clean and affordable energy is a major driver for prosperity and well-being of a people, the results from this study aim at contributing towards increased energy access and rapid electrification for Burkina Faso and other West African countries with similar conditions.

Credit Author Statement

Hamza Abid: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Writing – original draft. Jagruti Thakur: Methodology, Validation, Investigation, Writing – review & editing, Supervision, Project administration. Dilip Khatiwada: Methodology, Writing – review & editing. David Bauner: Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] World Bank. World Bank open data. ID: EG.ELC.ACCE.UR.ZS; 2017.
- [2] SE4all, "ECOWAS SE4ALL network".
- [3] Power Africa. Power Africa fact sheet. Burkina Faso; 2020.
- [4] Moner-Girona M, Bódis K, Korgo B. Mapping the least-cost option for rural electrification in Burkina Faso. 2017. <https://doi.org/10.2760/900097>.
- [5] Moner-Girona M, Bódis K, Huld T, Kougiass I, Szabó S. Universal access to electricity in Burkina Faso: scaling-up renewable energy technologies. *Environ Res Lett* 2016;11(8):1–15. <https://doi.org/10.1088/1748-9326/11/8/084010>.
- [6] Tractebel. Update of the ECOWAS revised master plan for the development of power generation; Priority Investment Program and Implementation Strategy. Brussels; 2018.
- [7] M. Bissiri, P. Moura, N. C. Figueiredo, and P. Pereira da Silva, "A geospatial approach towards defining cost-optimal electrification pathways in West Africa," *Energy*, vol. 200, 2020, doi: 10.1016/j.energy.2020.117471.
- [8] Clean Energy Wire. Journalism for energy transition. 2019.
- [9] Magazine PV. Burkina Faso announces plan to build eight solar parks. 2018.
- [10] West African Power Pool. ECOWAS master plan for the development of regional power generation and transmission infrastructure. *Africa Energy Forum*; 2019.
- [11] Bloomberg, "A behind the scenes take on lithium-ion battery prices."
- [12] World Bank. Accelerating battery storage for development. 2018.
- [13] Deane P, Gallachóir BÓ. Pumped hydro energy storage. *Handb. Clean Energy Syst*; 2015, p. 1–16. <https://doi.org/10.1002/9781118991978.hces137>.
- [14] IEA. Analysis from renewables. 2018.
- [15] IRENA. Renewable energy technologies: cost analysis series. *Green Energy Technol*. 2012;20(5):231–56. https://doi.org/10.1007/978-3-642-20951-2_8.
- [16] United Nations Industrial Development Organization. World small hydro-power development report. 2016. p. 1–44.
- [17] Vad Mathiesen B. Comparative analyses of seven technologies to facilitate the integration of fluctuating renewable energy sources. *IET Digit. Libr*. 2009;3(2): 190–204. <https://doi.org/10.1049/iet-rpg:20080049>.
- [18] Lund H, Andersen AN, Østergaard PA, Mathiesen BV, Connolly D. From electricity smart grids to smart energy systems e A market operation based approach and understanding Keywords: smart grid Smart energy systems Large-scale integration of wind and solar Electricity market Flexible Combined Heat and Power production (CHP). 2012. <https://doi.org/10.1016/j.energy.2012.04.003>.
- [19] Connolly D, Lund H, Mathiesen BV, Leahy M. A review of computer tools for analysing the integration of renewable energy into various energy systems. *Appl Energy* 2010;87(4):1059–82. <https://doi.org/10.1016/j.apenergy.2009.09.026>.
- [20] Trotter PA, McManus MC, Maconachie R. Electricity planning and implementation in sub-Saharan Africa: a systematic review. *Renew Sustain Energy Rev* 2017;74(June 2016):1189–209. <https://doi.org/10.1016/j.rser.2017.03.001>.
- [21] Sinha S, Chandel SS. Review of software tools for hybrid renewable energy systems. *Renewable and sustainable energy reviews*, vol. 32. Pergamon; Apr. 01, 2014. p. 192–205. <https://doi.org/10.1016/j.rser.2014.01.035>.
- [22] Yimen N, Hamandjoda O, Meva'a L, Ndzana B, Nganhon J. Analyzing of a photovoltaic/wind/biogas/pumped-hydro off-grid hybrid system for rural electrification in Sub-Saharan Africa - case study of Djoundé in Northern Cameroon. *Energies* 2018;11(10). <https://doi.org/10.3390/en11102644>.
- [23] Shrestha GB. A study on optimal sizing of stand-alone photovoltaic stations. *IEEE Trans Energy Convers* 1998;13(4):373–8. <https://doi.org/10.1109/60.736323>.
- [24] Chattopadhyay D, Frydas N, Kramskaya T, Tavoulareas E. Battery storage in developing countries: key issues to consider. *Electr J* 2019;32(2):1–6. <https://doi.org/10.1016/j.tej.2019.01.012>.
- [25] Nfah EM, Ngundam JM. Feasibility of pico-hydro and photovoltaic hybrid power systems for remote villages in Cameroon. *Renew Energy* 2009;34(6): 1445–50. <https://doi.org/10.1016/j.renene.2008.10.019>.
- [26] Adaramola MS, Agelin-Chaab M, Paul SS. Analysis of hybrid energy systems for application in southern Ghana. *Energy Convers Manag* 2014;88(2014): 284–95. <https://doi.org/10.1016/j.enconman.2014.08.029>.
- [27] A. S. Aziz, M. F. N. Tajuddin, M. R. Adzman, M. F. Mohammed, and M. A. M. Ramli, "Feasibility analysis of grid-connected and islanded operation of a solar PV microgrid system: a case study of Iraq," *Energy*, vol. 191, 2020, doi: 10.1016/j.energy.2019.116591.
- [28] Mensour ON, El Ghazzani B, Hlimi B, Ihlal A. A geographical information system-based multi-criteria method for the evaluation of solar farms locations: a case study in Souss-Massa area, southern Morocco. *Energy* 2019;182: 900–19. <https://doi.org/10.1016/j.energy.2019.06.063>.
- [29] Sigarchian SG, Paleta R, Malmquist A, Pina A. Feasibility study of using biogas engine as backup in a decentralized hybrid (PV/Wind/Battery) power generation system - case study Kenya. *Energy* 2015;90(2):1830–41.
- [30] Toopshekan A, Yousefi H, Astaraei FR. Technical, economic, and performance analysis of a hybrid energy system using a novel dispatch strategy. *Energy* 2020;213. <https://doi.org/10.1016/j.energy.2020.118850>.
- [31] Ma T, Yang H, Lu L. Feasibility study and economic analysis of pumped hydro storage and battery storage for a renewable energy powered island. *Energy Convers Manag Mar*. 2014;79:387–97. <https://doi.org/10.1016/j.enconman.2013.12.047>.
- [32] Yamegueu D, Azoumah Y, Py X. Experimental and economical study of sustainable electricity generation by solar PV/diesel hybrid systems without storage for off grid areas. *WIT Trans Ecol Environ* 2011;143:37–49. <https://doi.org/10.2495/ESUS110041>.
- [33] Connolly D, MacLaughlin S. Locating sites for pumped hydroelectric energy storage. 2011.
- [34] Huynen J. Underground pumped hydro storage: flatland large-scale electricity supply. 2018.
- [35] Adeoye O, Spataru C. Modelling and forecasting hourly electricity demand in West African countries. *Appl Energy* 2019;242:311–33. <https://doi.org/10.1016/j.apenergy.2019.03.057>.
- [36] Population des principales villes - Burkina Faso Open Data."
- [37] Lazard. Lazard's levelized cost of energy analysis 12.0. "; 2018.
- [38] Kenfack J, Neirac FP, Tatiéte TT, Mayer D, Fogue M, Lejeune A. Microhydro-PV-hybrid system: sizing a small hydro-PV-hybrid system for rural electrification in developing countries. *Renew Energy* 2009;34(10):2259–63. <https://doi.org/10.1016/j.renene.2008.12.038>.
- [39] Lazard. Lazard's levelized cost of storage analysis 4.0. 2018. <https://doi.org/10.1109/prime.2018.8430314>.
- [40] Barbour E, Wilson IAG, Radcliffe J, Ding Y, Li Y. A review of pumped hydro energy storage development in significant international electricity markets. *Renew Sustain Energy Rev* 2016;61:421–32. <https://doi.org/10.1016/j.rser.2016.04.019>.
- [41] Fu R, Feldman D, Margolis R. U.S. Solar photovoltaic system cost benchmark: Q1. 2018. <https://doi.org/10.7799/1325002>.
- [42] Dol AW, Chalise R, Hadjerioua B, Manwaring M, Bishop N. ORNL/TM-2016/590 development and implications of a predictive cost methodology for modular

- pumped storage hydropower (m-PSH) projects in the United States. 2011.
- [43] Lambert T, Gilman P, Lilienthal P. Micropower system modeling with homer. *Integr. Altern. Sources Energy* 2006;379–418. <https://doi.org/10.1002/0471755621.ch15>.
 - [44] HOMER, "Break even grid extension distance."
 - [45] Deichmann U, Meisner C, Murray S, Wheeler D. The economics of renewable energy expansion in rural Sub-Saharan Africa. *Energy Pol* 2011;39(1):215–27. <https://doi.org/10.1016/j.enpol.2010.09.034>.
 - [46] Ouedraogo BI, Kouame S, Azoumah Y, Yamegueu D. Incentives for rural off grid electrification in Burkina Faso using LCOE. *Renew Energy* 2015;78: 573–82. <https://doi.org/10.1016/j.renene.2015.01.044>.
 - [48] Bin L, Stocks M, Blakers A, Anderson K. Geographic information system algorithms to locate prospective sites for pumped hydro energy storage. 2018. <https://doi.org/10.1016/j.apenergy.2018.03.177>.
 - [49] Hansen K. Decision-making based on energy costs: comparing leveled cost of energy and energy system costs. *Energy Strateg. Rev.* 2019;24(December 2018):68–82. <https://doi.org/10.1016/j.esr.2019.02.003>.
 - [50] Fodhil F, Hamidat A, Nadjemi O. Potential, optimization and sensitivity analysis of photovoltaic-diesel-battery hybrid energy system for rural electrification in Algeria. *Energy* 2019;169:613–24. <https://doi.org/10.1016/j.energy.2018.12.049>.
 - [51] Clarke DP, Al-Abdeli YM, Kothapalli G. Multi-objective optimisation of renewable hybrid energy systems with desalination. *Energy* 2015;88:457–68. <https://doi.org/10.1016/j.energy.2015.05.065>.
 - [52] Ouedraogo NS. Energy consumption and economic growth: evidence from the economic community of West African States (ECOWAS). *Energy Econ* 2013;36:637–47. <https://doi.org/10.1016/j.eneco.2012.11.011>.
 - [53] IEA. Net metering and PV self-consumption in emerging countries - IEA-PVPS. 2018 [Online]. Available: <https://iea-pvps.org/key-topics/t9-net-metering-and-pvdevelopment-in-emerging-countries-en-report/>. [Accessed 25 January 2021].